Study Protocol v1.0 (final, for OSF registration)

From Fields to Dose: Evaluating Large Language Models

for Weight-Based Pediatric Medication Extraction and Dosing Consistency

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Purpose of this document. This is the study protocol timestamped on OSF before any evaluated model is run on the locked test set. Version 1.0 resolves every open verification and decision item of the September draft and records the final composition of the locked test set. Any later change is recorded in the deviations log (Section 21 and protocol/DEVIATIONS.md in the reproducibility package).

0. Registration metadata

Field	Value
Author	Vinod Rufus Motani (sole author). ORCID: https://orcid.org/0009-0005-3459-0859
Role	Independent researcher. This study is conducted on personal time, personal hardware, and personal accounts. No employer resources, systems, accounts, or data are used.
Data	Synthetic only. No real patient data, no protected health information, no human subjects. No IRB review required; this is stated in the paper.
Clinical review	None. No pharmacist or clinician participates. Ground truth is known by construction; clinical plausibility is validated against public references and automated audits (Section 8). The limitation is stated verbatim in the paper (Section 19).
Funding	Self-funded. Test-phase ceiling USD 200 in API usage (Section 10).
Conflicts	None declared.
Reporting standard	TRIPOD-LLM (Gallifant J, et al. Nature Medicine 2025;31:60 to 69. doi:10.1038/s41591-024-03425-5)
Registration	OSF Open-Ended Registration, with the locked test set hashes (SHA-256, LOCK.json, 3 October 2026): records 492d7ca2209719eb8bf96548403dab1e26cb9fed36be3482733dd69790cea7f4; scenarios 432be74fe65c2164ad9e21705821f5814055994f389526614aa744a671c220cb; repeat_subset f1012bdcb01f9cdcf80cf000b0ad6831352d9a98a6e57a0e8ebd207dbdfa4196; scoring code and prompts 7be9bc8704e09b94014741847f16b047b115c7b2c7c7846ec01afe03445461ed; n_records 361.

1. Research questions and hypotheses

Research questions

1. RQ1 (field level). How accurately do contemporary LLMs extract each element of a weight-based pediatric medication regimen from narrative synthetic records?
2. RQ2 (regimen level). How often are all required elements correct together, and how large is the gap between mean field accuracy and complete-regimen accuracy?
3. RQ3 (dosing consistency and consequence). When extracted values are used to recompute mg/kg/dose and mg/kg/day, how often does the result differ from the intended regimen by more than a pre-specified tolerance, and which error classes produce dose-consequential errors?
4. RQ4 (abstention). When required information is absent or conflicting, do models abstain appropriately or supply unsupported values?

5. RQ5 (architecture). Does a hybrid workflow (LLM interpretation plus deterministic arithmetic) reduce dose-consequential errors relative to LLM end-to-end reconstruction?
6. RQ6 (moderators, exploratory). How do documentation complexity and pediatric age band affect the above?

Pre-specified hypotheses

- H1. Complete-regimen accuracy will be at least 10 percentage points lower than mean field accuracy for every evaluated model.
- H2. The dose-consequential error rate will increase monotonically from complexity Level 1 to Level 3.
- H3. The hybrid workflow (C) will have a lower dose-consequential error rate than the end-to-end workflow (B) for the same model (primary comparison).
- H4. Dose-basis confusion (mg/kg/day vs mg/kg/dose) and weight-selection errors will together account for the largest share of dose-consequential errors.
- H5 (exploratory). Performance will be lower in the neonate and infant bands than in the adolescent band under matched challenges.

2. Design overview and tiering

Controlled benchmark study. Canonical regimens are generated first; synthetic narrative records are written from them; models extract a fixed structured representation; deterministic software reconstructs exposure; results are scored at four levels and stratified. Work is tiered so that the core is completed even if secondary items are cut.

Tier	Contents	Rule
Core (mandatory)	Canonical ground truth; dosing-source provenance; challenge matrix; three complexity levels; generator exclusion; dev/test lock; Levels 1 to 4 scoring; abstention scoring; age-band and complexity stratification; error taxonomy; workflows B and C; repeated runs on the stratified subset; clustered bootstrap; reproducibility package.	Must be complete before submission.
Secondary (time-boxed)	Workflow A rules-based baseline (maximum two working days); matched-challenge subgroup analysis.	Included if complete; otherwise reported as not done.
Exploratory	Medication-level analysis; generation-method analysis; mixed-effects models.	Appendix only; labeled exploratory; no hypothesis claims.
Excluded	Established biomedical IE systems as baselines; any clinician or pharmacist review; real-data validation.	Not part of this study. Real-data validation is named as future work.

3. Canonical ground truth

3.1 Canonical regimen schema

Every scenario is a structured object with the fields below. Fields marked (opt) are present only where the scenario requires them.

- scenario_id; drug (ingredient name as written on the FDA label; RxNorm concept identifiers were not retrieved); indication label (from the dosing source)

- age (days for neonates and infants, months and years otherwise); age_band (Section 6); sex (only where the growth reference requires it)
- current_weight_value; current_weight_unit (kg or lb as documented); weight_timestamp; prior_weight_value (opt); prior_weight_timestamp (opt); birth_weight (opt, neonates)
- dose_basis (mg/kg/dose | mg/kg/day | mcg/kg/dose | mcg/kg/day | fixed); weight_normalized_dose; dose_per_administration; dose_unit; frequency (per day, plus abbreviation as documented); route; formulation (opt); concentration (opt); administration_volume (opt); max_dose_per_dose (opt); max_dose_per_day (opt)
- order_status (active | discontinued | historical); order_timestamp; distractor_orders (opt, list of other medications with their own fields)
- expected_mg_per_kg_per_dose; expected_mg_per_kg_per_day (computed deterministically from the fields above)
- abstention_case (boolean); missing_fields (opt, list); conflict_fields (opt, list)

3.2 Deterministic computation

Expected exposure is computed by a single audited function: dose_per_administration = weight_kg × weight_normalized_dose (capped at max_dose_per_dose where present); daily exposure = dose_per_administration × frequency_per_day (capped at max_dose_per_day where present); mg/kg/day = daily exposure ÷ weight_kg. Units are normalized to mg and kg before any arithmetic. Where the scenario defines the regimen in mg/kg/day, the per-dose value is derived by division by frequency.

3.3 Validation before lock (no clinician)

7. Unit tests on every arithmetic and unit-conversion path, including lb to kg, mcg to mg, per-day to per-dose, and cap application.
8. Property tests: for every scenario, the narrative-to-canonical round trip (author's reference parser) must recover the canonical values exactly; any failure blocks lock.
9. Plausibility gates (Section 8.1) must pass for every non-abstention scenario.
10. Automated audit of every generated record by a model family not under test, using the rubric in Section 8.2; flagged records are corrected or removed before lock.
11. Blinded self-audit of a random 10% of records against the canonical values and the dosing-source table; every correction is logged and the correction rate is reported.

Known-by-construction ground truth removes annotation error. It does not remove implementation error. Steps 1 to 5 exist to catch implementation error, and their results are reported in the paper.

4. Dosing-source hierarchy and provenance

No dosing pattern enters the benchmark without a public, versioned source. Sources are used in the following order; a scenario is excluded if no source in the hierarchy supports it.

12. FDA-approved product labeling (Dosage and Administration; Pediatric Use), retrieved from DailyMed or the openFDA drug label API. Both expose the same SPL corpus; DailyMed set ID and version are recorded.
13. WHO Model Formulary for Children (2010) and the WHO Model List of Essential Medicines for Children (9th list, 2023) were planned as cross-checks of age appropriateness and baseline ranges. At lock, every dosing row is sourced from a current FDA label (SPL set-id and version recorded per row in sources/dosing_sources.csv) and the WHO cross-check was not performed; this is recorded as a deviation.

14. No national pediatric formulary (Kinderformularium, SwissPedDose) was used: FDA labeling alone supported the weight-based pattern for all 21 sourced drugs, so no secondary source was needed (recorded as a deviation from the draft's optional cross-check).

Provenance record per drug and dosing pattern: source; set ID or document ID; version or publication date; retrieval date; indication; age restriction; weight-normalized dose range; maximum per dose and per day where stated; route and formulation; notes. The table is released with the package. Proprietary references (Lexicomp, BNF for Children, Harriet Lane, NeoFax) are not used and are named as excluded.

5. Drug selection

Target: 15 to 20 drugs. Inclusion: commonly used in pediatrics; weight-based dosing stated in a source from Section 4; oral or parenteral; at least one formulation with a public concentration. Exclusion: no public weight-based dosing source; dosing dependent on renal function, serum levels, or body surface area in the primary scenario; any drug whose only public dosing is for a condition outside the scenario set.

Composition constraints (to make the challenge matrix testable): at least 3 drugs dosed in mcg/kg; at least 5 drugs with a labeled maximum per dose or per day; at least 3 oral liquids requiring concentration-to-volume conversion; at least 4 parenteral drugs; at least 2 drugs commonly written both per dose and per day.

Final list fixed at lock (21 drugs sourced from FDA labels, 20 represented in the locked test set because one drug yielded no scenario inside its labeled age band at the sampled ages; see sources/dosing_sources.csv for the label set-id, version and pattern of each): amoxicillin; amoxicillin-clavulanate; cephalexin; cefdinir; azithromycin; clindamycin; fluconazole; ampicillin; ceftriaxone; gentamicin (per-dose and per-day rows); vancomycin; acetaminophen (intravenous); ibuprofen; ondansetron (intravenous); dexamethasone; prednisolone; famotidine; levothyroxine (mcg); digoxin (mcg); fentanyl (mcg); midazolam. Compared with the 23 September candidate list, clonidine, morphine and enoxaparin were replaced by clindamycin, fluconazole, digoxin and fentanyl because their current labels state explicit pediatric weight-based regimens.

6. Pediatric age bands and weights

Age bands follow the NICHD pediatric terminology standard (Williams K, Thomson D, Seto I, et al. Standard 6: Age groups for pediatric trials. Pediatrics 2012;129(Suppl 3):S153, S160. doi:10.1542/peds.2012-0055I; PubMed 22661762): term neonate 0 to 27 days; infancy 28 days to 12 months; toddler 13 months to 2 years; early childhood 2 to 5 years; middle childhood 6 to 11 years; adolescence 12 to 17 years (the standard's early adolescence, truncated at the 18th birthday). As implemented (sources/age_bands.json): neonate 0 to 27 days; infant 28 to 365 days; toddler and young child 366 to 2190 days (toddler and early childhood combined); school-age child 2191 to 4380 days; adolescent 4381 to 6574 days. Scenarios are generated only in bands the drug's FDA label supports (Section 4), so some drug × band cells are empty by design.

Weights are sampled from growth-reference LMS parameters by age and sex: WHO Child Growth Standards weight-for-age tables (boys and girls, 0 to 24 months; https://www.who.int/tools/child-growth-standards/standards/weight-for-age) for children under 2 years, and the CDC 2000 growth charts weight-for-age LMS file for ages 2 to 20 years (https://www.cdc.gov/growthcharts/data/zscore/wtage.csv). Both files were downloaded by hand on 2 October 2026 (scripted download was refused by the servers) and are stored unmodified in sources/ with their retrieval date in protocol/DEVIATIONS.md. Sampling uses z in [-2, +2] for 90% of scenarios and a pre-specified 10% in the tails with z in ±[2.05, 2.5], with weight additionally capped at 2.2 times the age-sex median because the LMS transform is unreliable beyond that range. Neonatal scenarios include a birth weight distinct from the current weight in at least half of cases.

7. Synthetic record generation

7.1 Generation methods

- Deterministic templates written by the author (pediatric visit, admission, discharge, and medication-reconciliation note styles).
- Controlled paraphrase of template output by one generator model. Generator exclusion rule: the generator's model family is excluded from the set of evaluated models. The generator and its version are recorded.
- Every record carries the generation method as metadata so results can be examined by method (exploratory).

7.2 Complexity levels (fixed at lock)

- Level 1, explicit: single current weight, single active order, unambiguous units and frequency, no distractors.
- Level 2, naturalistic: same regimen inside realistic narrative with vitals, history, and at least one other medication; standard abbreviations (BID, TID, QID, q6h, q8h); at least one non-dosing numeric value near the weight.
- Level 3, challenging: at least two of: multiple weights (prior or birth weight present), lb documentation, mcg drug, per-day order requiring per-dose derivation, discontinued order alongside the active one, concentration requiring volume computation, dose change within the record, or a conflicting value.

7.3 Challenge matrix

Challenge	Purpose	Detected error class
Multiple weights; historical vs current	Weight selection under temporal cues	Weight-selection; temporal
Pounds documented, kg expected	Unit reasoning	Weight-unit
mcg-dosed drug	Scale reasoning	Unit-scale
mg/kg/day vs mg/kg/dose	Dose-basis reasoning	Dose-basis
Labeled maximum reached	Cap application	Dose-cap
Concentration and volume present	Strength vs dose vs volume	Strength-vs-dose; volume-vs-dose
Discontinued and active orders together	Order status reasoning	Temporal; association
Multiple medications	Association to the correct order	Association
Abbreviated frequency and route	Normalization	Frequency; route; normalization
Missing weight or frequency; conflicting values	Abstention behavior	Unsupported-value hallucination; inappropriate abstention

7.4 Abstention cases

A fixed 15% of test records are abstention cases with a required field missing or two irreconcilable values. Canonical ground truth for those fields is null, and the correct model behavior is an explicit insufficient-information output for the affected field and for any computed quantity that depends on it.

8. Realism and plausibility validation (no clinician)

8.1 Plausibility gates (hard, applied at generation)

- Weight z-score within the band defined in Section 6 for the scenario's age and sex.
- Weight-normalized dose within the source's stated range for the indication and age; maxima applied where stated.
- Frequency valid for the route and formulation per the source; concentration and volume consistent with dose.
- Order timestamps consistent with weight timestamps; discontinued orders dated before active ones.

8.2 Automated audit (soft, reported)

Every record is reviewed by a model family not under test against a fixed rubric: reads as a plausible pediatric note; the intended regimen is recoverable by a careful reader; no internal contradiction other than those planted by design; no protected information; no identifiers. Flag rate, corrections, and removals are reported.

8.3 Style anchoring (reported)

Readability, note length, section structure and abbreviation density may be compared descriptively against publicly available de-identified note samples; if performed, the source and its licence are recorded in the package. This is not a registered outcome and no claim of equivalence is made.

8.4 Blinded self-audit

As in Section 3.3, step 5. Correction rate reported.

9. Development set and locked test set

- Split at the canonical-scenario level. No scenario contributes records to both sets. Development set as built: 172 records from 172 scenarios (one variant per scenario, 50% paraphrased by the generator model). Test set: Section 14.
- Prompt engineering, schema changes, normalization rules, retry policy, and malformed-output handling are developed only on the development set.
- At lock: the prompt, schema, scoring code, tolerance definitions, and analysis plan are frozen; a cryptographic hash of the test set and of the scoring code is recorded in the OSF registration.
- After the test set is opened, no prompt or scoring change is permitted. Any unavoidable change is recorded as a deviation with its reason and the affected analyses are labeled.

10. Models

- Final list, fixed at lock (3 October 2026), with pinned identifiers as in config/models.yaml: (1) gpt-5.5-2026-04-23 (OpenAI; dated snapshot; leading commercial model, provider 1); (2) claude-sonnet-5 (Anthropic; leading commercial model, provider 2); (3) gemini-3.1-pro-preview (Google, via the OpenAI-compatible endpoint; third commercial provider); (4) deepseek-v4-pro (DeepSeek; open-weight family accessed through its API); (5) llama3.1:8b (Meta Llama 3.1 8B, run locally through Ollama; the small, low-cost open-weight

model). The generator and realism auditor are qwen2.5:14b (Alibaba Qwen family, local), which is not evaluated; the runner enforces the exclusion. Identifiers are the same snapshots used in the author's companion benchmark registration (pinned 25 September 2026) so the two studies are comparable.

- Access rule: API only with pinned model identifiers. No consumer web interfaces.
- Recorded per model: exact identifier and snapshot; provider; evaluation dates; temperature (0 unless unsupported); seed where supported; reasoning or inference settings; system and task prompts; structured-output settings; retry and repair rules; tool availability (none); any pre- or post-processing; cost per record.
- Budget ceiling: USD 200 in API usage for the test phase (development-phase spend is excluded and is reported separately in the run manifest). If the ceiling would be exceeded, the smallest commercial model is dropped first, then the repeated-run subset is reduced to the minimum in Section 15, and the change is logged.

11. Workflows and output contract

- Workflow A (secondary, time-boxed): rules-based extraction using regular expressions and a small grammar for weights, doses, units, frequencies, and routes, followed by the deterministic calculator. Two working days maximum.
- Workflow B (core): LLM end-to-end. The model returns the structured fields and the computed mg/kg/dose and mg/kg/day.
- Workflow C (core): hybrid. The model returns only the structured fields; the deterministic calculator computes exposure and consistency.
- Output contract: one JSON object per record following the schema in Section 3.1, with explicit null and an insufficient_information flag per field. One repair retry is allowed for malformed JSON; a second failure is scored as a complete-regimen failure and counted separately.

12. Evaluation levels and scoring rules

Level 1: field accuracy

Each required field is scored correct or incorrect after normalization: units mapped to canonical units; numeric values compared within 1% relative tolerance after unit normalization; frequency mapped to times per day; route and drug matched by canonical code. Missing required field counts as incorrect except in abstention cases.

Level 2: complete-regimen accuracy

All required fields for the scenario correct together. The required-field set is defined per scenario at generation and frozen.

Level 3: dosing consistency

The deterministic calculator recomputes mg/kg/dose and mg/kg/day from the model's extracted fields (Workflow C) or reads the model's own computed values (Workflow B). A record is dosing-consistent if both recomputed values are within the Level 4 tolerance of the canonical values.

Level 4: dose-consequence

Primary definition: an error is dose-consequential if the absolute relative deviation of recomputed mg/kg/dose or mg/kg/day from the canonical value exceeds 10%. Sensitivity analysis at 20%. A separate binary flag records any recomputed exposure that exceeds the labeled maximum. The 10% threshold follows the modified-Delphi

definition of pediatric dosing error (Roumeliotis N, Pullenayegum E, Rochon P, et al. A modified Delphi to define drug dosing errors in pediatric critical care. BMC Pediatrics 2020;20:488. doi:10.1186/s12887-020-02384-3), in which expert clinicians agreed that dosing more than 10% above or below the reference range constitutes an error for most intermittent medications. No claim about patient harm is made from this definition.

Abstention scoring

Per abstention case: appropriate abstention (correct field flagged insufficient); unnecessary abstention (flag on a field that was present); unsupported-value hallucination (a value supplied for a missing field); incorrect inference (a computed quantity supplied despite a missing dependency).

Primary endpoint

Complete dose-consistent regimen accuracy: the proportion of non-abstention test records in which all required fields are correct and the recomputed exposure is within the primary tolerance. Abstention cases have their own primary metric: appropriate-abstention rate.

13. Error taxonomy and coding

Classes: extraction; normalization; association; temporal; weight-selection; weight-unit; unit-scale (mg vs mcg); dose-basis; strength-vs-dose; volume-vs-dose; frequency; route; dose-cap; adult fixed-dose assumption; unsupported-value hallucination; arithmetic inconsistency. Each error is also tagged dose-consequential or not under Section 12.

Coding procedure: classes that can be determined by comparing the model output with the canonical record (weight-selection, weight-unit, unit-scale, dose-basis, frequency, route, dose-cap, temporal, hallucination, arithmetic) are assigned automatically by rules committed before lock. Residual errors are coded manually by the author using a written codebook, blinded to model identity, in randomized order. The automatic versus manual fraction is reported. A random 10% of manual codes are re-coded after a two-week interval and intra-rater agreement is reported.

14. Sample size

Test set as locked on 3 October 2026 (LOCK.json hashes in Section 0): 361 records from 361 canonical scenarios covering 20 drugs (one of the 21 sourced drugs produced no in-label scenario), 5 reporting bands and 3 complexity levels, two variants per drug × band × complexity cell where the label supports the band, approximately 15% abstention cases, approximately 32% of records paraphrased by the generator model. Each record is its own scenario, so the clustering unit for the bootstrap is the record (361 clusters). The 23 September draft estimated 600 to 900 records; the label-derived age restrictions (Section 4) excluded 267 candidate scenarios, and the generator was not re-expanded after the self-audit so that the audited set is the locked set. With 361 clusters and a primary-endpoint proportion near 0.5, the clustered bootstrap 95% confidence interval half-width is approximately 5 percentage points, which remains sufficient to detect a 10-point difference for the primary comparison. Two level-3 neonatal records with timeline defects were removed before lock (protocol/DEVIATIONS.md). Development set: 172 records from 172 separate scenarios.

15. Repeated runs and stability

Primary results use the lowest-variance configuration each model supports (temperature 0 where the API accepts it, fixed seed where available; GPT-5.5 does not accept a fixed temperature and is run at the provider default,

recorded in the manifest). Because API models are not deterministic, every model is run three times on a stratified random subset of 108 test records (30%, stratified by band and complexity; data/test/repeat_subset.txt, hash in Section 0). Reported: field-level agreement across runs, complete-regimen stability, dosing-consistency stability, and the proportion of records whose dose-consequence classification changes between runs. The subset and run count were fixed at lock, before the test set was opened.

16. Statistical analysis plan

- Unit of analysis: record; clustering: canonical scenario. All confidence intervals use clustered bootstrap resampling of scenarios (2,000 resamples).
- Primary comparison (H3): Workflow C minus Workflow B on the primary endpoint, per model, paired at the record level, with clustered bootstrap 95% CI. The comparison is declared supported if the CI excludes zero in the hypothesized direction for the majority of evaluated models.
- Secondary comparisons between models on the primary endpoint use the same paired clustered bootstrap with Holm correction across the pre-specified set of pairwise contrasts.
- H1: report mean field accuracy minus complete-regimen accuracy per model with CI.
- H2 and H5: report the endpoint by complexity level and by band with CIs; exploratory mixed-effects logistic regression with a scenario random intercept and fixed effects for model, band, complexity, and generation method; results labeled exploratory.
- H4: report dose-consequential errors by class as proportions with CIs.
- Abstention metrics reported as proportions with CIs; no formal comparison.
- Reporting follows TRIPOD-LLM. All analyses are run by a single committed script; no result is reported that the script does not produce.

17. Reproducibility package

- Repository (public at submission): protocol; canonical scenarios and generator; dosing-source provenance table; templates; generated records (development and test); schema; prompts; model configurations; scoring code; deterministic calculator with tests; error codebook and coded errors; analysis script; results tables and figures; cost and run logs.
- Licenses: data and documents CC-BY-4.0; code MIT. Model versions and evaluation dates pinned in a single file.
- Repository structure: protocol/, sources/, generator/, data/dev/, data/test/, harness/, scoring/, analysis/, results/, README.

18. Timeline and budget (solo)

The study is self-funded and conducted on personal time. The test-phase spending ceiling and the rule applied if it is exceeded are given in Section 10; actual API spend is reported in the run manifest released with the package.

19. Limitations statement (verbatim for the paper)

This study uses synthetic pediatric records generated from canonical regimens; ground truth is known by construction and no clinician or pharmacist reviewed the records. Clinical plausibility was enforced with automated gates against public dosing sources and growth references and audited by a model family not under evaluation; these procedures reduce but do not eliminate the possibility of unrealistic or implementation-derived cases, and the correction rates from validation are reported. Results characterize model behavior on controlled synthetic text and do not establish performance on real clinical documentation, clinical safety, patient outcomes, or readiness for deployment. Real-data validation under appropriate governance is required before any clinical use. The realism auditor and the paraphrase generator belong to the same model family; the independent check on the ground truth is the author's blinded read of a 10% sample, whose correction rate is reported. Because leading models achieved near perfect field accuracy on the development set, ceiling effects on the primary endpoint are a possible outcome for those models; the pre-specified strata where discrimination is expected are complexity level 3, the abstention and conflict cases, and the small open-weight model, and a null result for H3 in a model at ceiling will be reported as such rather than reinterpreted.

20. Ethics and integrity

- No human subjects; synthetic data only; no identifiers; no protected health information.
- No employer involvement: no employer data, systems, accounts, hardware, time, or approval was used or required. The author's affiliation is stated as independent researcher.
- Sole authorship reflects that conception, design, implementation, analysis, and writing were performed by the author. No other person contributed. Generative AI assistants were used as software development and drafting aids under the author's direction; every design decision, dosing source, line of scoring logic, and sentence of this protocol was reviewed and is the author's responsibility. The assistants used for development are not counted as evaluated models, and the one evaluated model family that overlaps with them is run only through the pinned API identifier recorded in Section 10.
- No claim of clinical safety, superiority to professionals, or real-world effectiveness will be made.

21. Deviations log and version history

Any change after registration is recorded here with date, section, original text, new text, reason, and the analyses affected. The registered protocol version and the final version are both released with the package.

Version	Date	Change
1.0	2026-09-23	Initial pre-registration draft
1.0 final	2026-10-03	Resolved all open verification and decision items: NICHD age-band citation and implemented boundaries; WHO/CDC LMS file sources; final five-model list with reasons; budget ceiling USD 200; TRIPOD-LLM citation. Recorded the locked test set (361 records/scenarios, 108-record repeat subset) and development set (172) as built, replacing the draft estimate of 600 to 900. Pre-

		lock development-phase changes (prompt schema example, scorer handling of non-scalar values and withheld per-dose amounts, billing-error handling, generator fixes for tail weights, neonatal timelines and age-appropriate distractors) are itemised in protocol/DEVIATIONS.md.

22. Key references for the related-work section

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